

Scalar product

Key Points

The scalar/dot product is

$$\mathbf{a} \cdot \mathbf{b} = a_1b_1 + a_2b_2 + a_3b_3$$

The angle between two vectors is given by

$$\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}||\mathbf{b}|}$$

So perpendicular vectors have a scalar product of zero

Basic Skills

Q1. Evaluate the following scalar products

a) $\begin{pmatrix} 5 \\ 3 \\ 6 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}$ **b)** $\begin{pmatrix} -1 \\ 0 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} -4 \\ 2 \\ -1 \end{pmatrix}$ **c)** $\begin{pmatrix} 3 \\ 1 \\ -4 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -2 \\ -1 \end{pmatrix}$

Q2. Find the magnitude of the following vectors

a) $\mathbf{a} = \begin{pmatrix} 3 \\ 4 \\ 5 \end{pmatrix}$ **b)** $\mathbf{b} = \begin{pmatrix} -1 \\ 2 \\ -5 \end{pmatrix}$ **c)** $\mathbf{c} = \begin{pmatrix} -1 \\ -2 \\ -5 \end{pmatrix}$

Q3. Find the angle between the following pairs of vectors

a) $\begin{pmatrix} 3 \\ 4 \\ 5 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ 1 \\ 8 \end{pmatrix}$ **b)** $\begin{pmatrix} 5 \\ -1 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} 10 \\ -2 \\ 4 \end{pmatrix}$

Competency Skills

Q4. Find the angle the coordinate (4,-2,3) makes with the y-axis

Q5. Given that the vectors $\mathbf{a} = \begin{pmatrix} x \\ 5 \\ 8 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 2 \\ 15 \\ x \end{pmatrix}$ are perpendicular find the value of x

Q6. Find the angles inside the triangle with coordinates (1,2,3), (5,-2,1) and (10,5,2)

Q7. Find the obtuse angle between the vectors $\mathbf{i} + \mathbf{j} + \mathbf{k}$ and $2\mathbf{i} + \mathbf{j} + 4\mathbf{k}$

Q8. Write down the value of $\mathbf{i} \cdot \mathbf{i} + 3\mathbf{j} \cdot \mathbf{k} + 2\mathbf{k} \cdot 4\mathbf{k}$

Q9. Explain why the scalar product is commutative ($\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}$)

Q10. Explain why $|\mathbf{a}|^2 = \mathbf{a} \cdot \mathbf{a}$

Advanced Skills

Q11. a) Draw a generic picture with two vectors from the origin $\mathbf{a}$ and $\mathbf{b}$ in 2D

b) State the subtraction angle formula for $\cos(A - B)$

c) By using the diagram and the subtraction angle formula prove that $\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}||\mathbf{b}|}$ in 2D

Q12. a) Prove that the scalar product is distributive ($(\mathbf{a} + \mathbf{b}) \cdot \mathbf{c} = \mathbf{a} \cdot \mathbf{c} + \mathbf{b} \cdot \mathbf{c}$)

b) Hence solve the vector equation $(\mathbf{a} + \mathbf{b}) \cdot \mathbf{b} = 0$ given $|\mathbf{a}| = 2$, $|\mathbf{b}| = 5$ and the angle between the vectors is 60°

Extension

Q13. Show that a projection of $\mathbf{a}$ into $\mathbf{b}$ (vector $\mathbf{a}$ aligned parallel and perpendicular to vector $\mathbf{b}$) we have the new vector given by $\frac{|\mathbf{a}|}{|\mathbf{b}|} \mathbf{b} \cos \theta$

Hint: Draw a right angled triangle with vector $\mathbf{a}$ as the hypotenuse